

Classifying Knowledge Domains of Theoretical and Quantitative Geography publications

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Session: *Quelle place pour une géographie théorique et quantitative critique ?*

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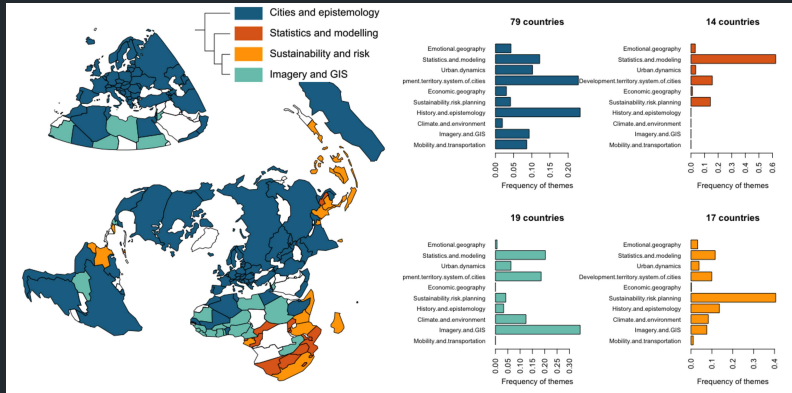
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ENSG
Géomatique

ÉCOLE NATIONALE
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GÉOGRAPHIQUES

Quantitative epistemology for reflexivity



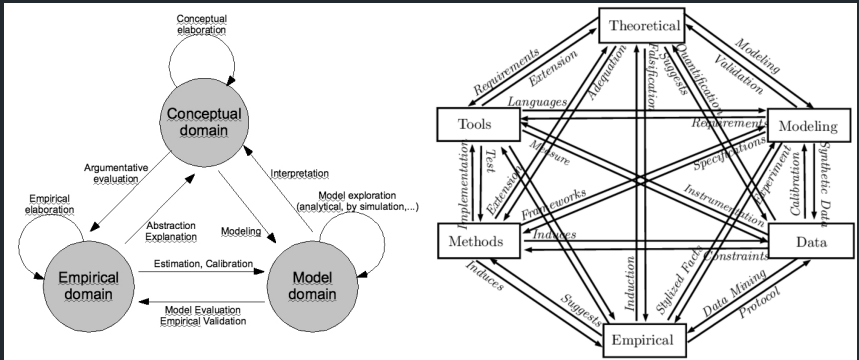
*Spatialised bibliometrics as a tool for a more reflexive and open science:
Cybergeonetworks project [Raimbault et al., 2021].*

<https://analytics.huma-num.fr/geographie-cites/cybergeonetworks/>

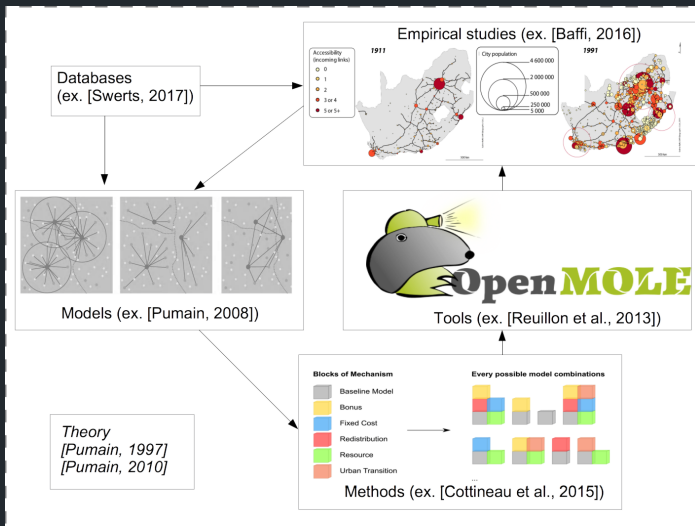
Cybergeo 30 anniversary on March 13th:

<https://cybergeo2026.sciencesconf.org/>

- Dialogue between theory and modelling already coined by D. Harvey in *Explanation in Geography* [Harvey, 1969].
- Core role of modelling and simulation to build theories, such as Pumain's Evolutionary Urban Theory [Pumain and Raimbault, 2020]
- Current need for redefining “transversal knowledge objects” and the position of TQG [Genre-Grandpierre, 2025]



(Left) Knowledge framework for agent-based modelling by [Livet et al., 2010]; (Right) Refinement with more Knowledge Domains (KDs) by [Raimbault, 2017a].



- Previous work on Pumain's Evolutionary Urban Theory [Pumain, 2018] by [Raimbault, 2017a] suggested an integration of KDs ; confirmed on larger corpuses (EUT and Zipf's law studies) by [Raimbault, 2025].
- More general ongoing epistemological research on TQG: [Raimbault, 2017b] (ECTQG 2017), [Raimbault, 2019] (ECTQG 2019), [Raimbault, 2023] (ECTQG 2023).
- Need to overcome limitations of manual labelling for a systematic mapping of TQG.

Research objective:

Evaluate machine learning methods for an automatic classification of KDs in TQG publications.

- corpus of TheoQuant proceedings (1993-2011, every two years)
- manual labelling of 234 abstracts; multi-class labelling
- translation to English using google translate API
- four knowledge domains to have enough abstracts in each class : Empirical (125), Theoretical (40), Modelling (56), Methodological (78)

Baseline: TF-IDF + Logit classifier

- **representation:** sparse vector of term frequencies
- **focus:** keyword importance
- **pros:** transparent, fast, low data requirement
- **cons:** ignores word order and polysemy

Deep Learning: SciBERT [Beltagy et al., 2019]

- **representation:** pre-trained dense contextual embeddings
- **focus:** Domain-specific semantics from 1.14M papers
- **pros:** Captures methodological nuance and context
- **cons:** Computationally expensive, risk of overfitting on small N

Training Protocol: 5-Fold Cross-Validation

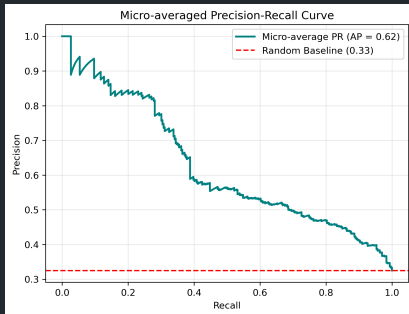
Parameters for SciBERT: 10 epochs, batch size 16, learning rate 2e-5, max seq length 256; local run without GPU

- **Backbone:** Based on the **BERT-base** architecture (Bidirectional Encoder Representations from Transformers)
 - 12 layers (Transformer blocks)
 - 768 hidden units and 12 attention heads; **110M** parameters
 - links tokens to their context (sentence), in a bidirectional way
- **SciBERT for science:** [Beltagy et al., 2019]
 - **Vocabulary (SciVocab):** Custom WordPiece vocabulary built from scientific papers (better handling of specific terms like *stochastic* or *model*)
 - **Pre-training Corpus:** 1.14 million papers from Semantic Scholar
- **Application to TQG (Fine-tuning):**
 - Input: Title and Abstract text → Output: One of 4 KDs
 - Training of the classifier via the **[CLS]** token embedding and a final linear layer for classification

Results: unoptimised classifiers

Unoptimised classifiers (probability threshold) do not classify correctly: same performance of random baseline, logit and scibert (F-score 0.48 ± 0.05 against 0.46 ± 0.03 for the baseline)

But models do extract some information:



Top terms used by logit: **Empirical:** urban, changes; **Theoretical:** concepts, geography; **Modelling:** model, simulation; **Methodological:** variables, method

→ select optimal threshold to classify from predicted probability

Model	Accuracy	Micro F1	Macro F1
Zero-R (Baseline)	0.252 (± 0.06)	0.472 (± 0.06)	0.175 (± 0.02)
TF-IDF + Logit	0.274 (± 0.06)	0.591 (± 0.02)	0.405 (± 0.08)
SciBERT	0.509 (± 0.05)	0.732 (± 0.02)	0.694 (± 0.04)

Accuracy: multi-class exact accuracy; Macro F1: average of F1 scores across all classes; Micro F1: global F1 with sample-level TPs/FPs/FNs

→ significant improvement on all indicators

→ with 0.5 accuracy, can be used as a pre-filtering tool but not yet as an automatic classification tool for systematic mapping

→ single-class classification, on the Evolutionary Urban Theory corpus ($N = 1252$) annotated by [Raimbault, 2025]

Model	Accuracy	Macro F1	Weighted F1
Zero-R (Baseline)	0.393 (± 0.03)	0.141 (± 0.01)	0.223 (± 0.03)
TF-IDF + Logit	0.650 (± 0.03)	0.594 (± 0.03)	0.650 (± 0.03)
SciBERT	0.798 (± 0.01)	0.743 (± 0.03)	0.796 (± 0.01)

→ again significant improvement on all indicators

→ **80% accuracy**, can be used reasonably as a systematic labelling tool

First results:

- Using a context-aware embedding for scientific texts (SciBERT) for classification **significantly improves classification**
- For multi-class classification, an **accuracy of 0.5** suggests its use as a **pre-filtering tool** only; for single-class, an **accuracy of 0.8** suggests a possible systematic mapping
- Promising application of **deep learning to mapping TQG**

Perspectives:

- SHAP [Mosca et al., 2022], hyperparameter optim., GPU training
- Test on larger corpuses (e.g. Zipf corpus by [Raimbault, 2025], $N = 858$) and full texts
- Application to other tasks: content extraction (model structure [Le Néchet and Raimbault, 2026]), level of evidence, complexity, ...

Open code and data:

<https://github.com/JusteRaimbault/GeoTheoQuantIntegration>

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